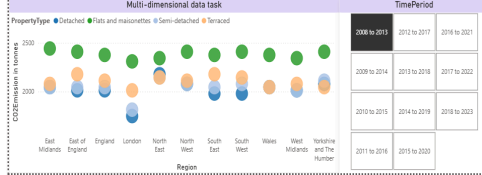
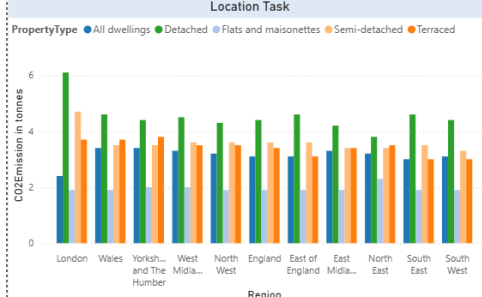


Coursework Description Sheet

Name: Akshay Bedre

Question	Description	Figure
Fit to Task/User needs		
Location task - How does the visualisation allow users to access the spread of carbon dioxide emission across the UK based on the property type?	In this assignment, I used a clustered column chart to show how the CO₂ emissions are distributed in different regions of the UK. These regions (London, Wales and North East) are displayed on the X-axis, whereas the CO₂ emissions (in tonnes) are presented on the Y-axis. The columns in the colour palette of any area are indicators of the type of property, such as detached, semi-detached, terraced, or flats and maisonettes. By using this visualisation, users can easily compare the level of carbon emissions of each type of property per region and can further see what a given property type makes the biggest or the smallest contribution of carbon emission in the UK.	Location Task PropertyType: All dwellings, Detached, Flats and maisonettes, Semi-detached, Terraced CO2Emission in tonnes Region: London, Wales, Yorksh... and The Humber, West Midlands, North West, England, East of England, East Midlands, North East, South East, South West
Time task - How does the visualization allow user to understand the evolution of energy efficiency based on the property type, and location?	I employed line chart to illustrate the change of the energy efficiency over time. The X-axis represents the time period whereas the Y-axis represents the total values of energy efficiency. The lines are associated with various types of property, and it is possible to distinguish the trends by the help of various colours. The graph has a clear description of the changes in efficiency of the energy over the time period and it outlines growth or decline tendencies of each property type. I also incorporated a region slicer, which can help the users to filter the data by region. Indicatively, with reference to all regions, flats have a higher average energy efficiency than other forms of property.	Time Task PropertyType: Detached, Flats and maisonettes, Semi-detached, Terraced Sum of EnergyEfficiency Region: England, East Midlands, East of England, London, North East, North West, South East

- Data Visualization

Multi-dimensional data task - How does the visualization allow user to identify correlation amongst at least three of the following parameters: property type, tenure, location, energy efficiency, and carbon dioxide emission?	In this assignment, I used a bubble (scatter) chart to examine the correlation between several parameters. The X-axis is the CO₂ emissions (in tonnes), whereas the Y-axis is the various regions. The colour of each bubble represents the type of property, which assists in visualisation of data groups. A time-period slicer is also included, which helps the user to discover how the emissions vary with varying time periods. It is an interactive system, and it enables users to see possible property type, region and CO₂ emissions correlations in real-time, so that the users can find the trends and comparisons in the most efficient way.	
Visualisation Principles		
Use of colour - How does the use of colour in this dashboard enhance the readability and effectiveness of the data presentation?	The colour scheme is well chosen and made easily understandable. Each type of property receives different and equalised colours in order to create a better user interface and make the data easy to contrast. I have also selected colours that will still be identifiable by users with colour vision deficiencies (e.g., deuteranopia) and enhance the inclusiveness and legibility of the dashboard	
Use of graphic design principles -How does the application of graphic design principles enhance the clarity and effectiveness of the data presentation in this dashboard?	The dashboard is well aligned, spaced and hierarchically organized. The charts are organized in a logical manner with uniform margins and labeled headings so that one can see the visuals without clutter. There is a visual hierarchy that is ensured by the use of bigger fonts in titles and bold section labels that enable users to move between tasks easily. The design is balanced between beauty and functionality which increases general clarity and understanding.	

- Data Visualization

Use of interaction - How does the use of interactive design elements improve the user's ability to explore and interpret data on this dashboard?	The dashboard is made interactive and exploratory with the use of interactive design features, e.g., slicers and dynamic KPIs. As an example, the time and region slicers allow the users to filter the data by the time or region, and the visuals are updated immediately. KPIs have average energy efficiency and average CO 2 emission (2018-2023), which give users a no-book insight. The line chart is dynamically updated regarding the selections of regions, which enables a natural comparison of various regions. Also the slicer used for the region is a drop down which makes it minimalist to see on the first go and use the slicer using drop down as needed	TimePeriod <table border="1"> <tr> <td>2008 to 2013</td><td>2012 to 2017</td><td>2016 to 2021</td></tr> <tr> <td>2009 to 2014</td><td>2013 to 2018</td><td>2017 to 2022</td></tr> <tr> <td>2010 to 2015</td><td>2014 to 2019</td><td>2018 to 2023</td></tr> <tr> <td>2011 to 2016</td><td>2015 to 2020</td><td></td></tr> </table>	2008 to 2013	2012 to 2017	2016 to 2021	2009 to 2014	2013 to 2018	2017 to 2022	2010 to 2015	2014 to 2019	2018 to 2023	2011 to 2016	2015 to 2020	
2008 to 2013	2012 to 2017	2016 to 2021												
2009 to 2014	2013 to 2018	2017 to 2022												
2010 to 2015	2014 to 2019	2018 to 2023												
2011 to 2016	2015 to 2020													
Use of text and legend - How do the use of text and legends contribute to the clarity and user comprehension of the data presented in this dashboard?	There is good usage of text and legends to enhance clarity and comprehension. The titles are all aligned to the center and are very brief helping to understand what each chart represents. Legends are put in visible positions with proper colour coding to each type of property so that the viewer can understand the data easily without any confusion. The low and restrained use of text does not create distractions, as the main points remain in the focus.	Multi-dimensional data task												

References

Add the list of references used in the description in the format of your choice.